

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – TECHNICAL PRESENTATION

Course Code:	CSA161b	Course Title:	Data Warehousing and Data Mining
CO Assessed:	CO1 – Precision (BL2, BL3, BL4)	Assessment:	Assessment Tool 2 – Technical Presentation
Weightage:	20% of CIA	Total Marks:	100 (2 Question × 100 Marks)
CO1 Statement:	<i>Design data warehouse architectures with appropriate dimensional models for effective decision support systems. BL1, BL2</i>		
Student Name:	M. Charan	Reg. No.:	192324115
Section / Batch:	CSE (A1 & D5)	Date:	19-7-26

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Decision support system, Operational versus Decision support System, Architecture of data warehouse, ETL, OLAP and data cubes, Dimensional data modelling -star snowflake schemas. Query Processing, Open-source Business Intelligence Tools. Develop an application to perform OLAP operations - Introduction to KNIME tool for reporting.	Q1, Q2	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Topic 1: Decision Support Systems (DSS)

☞ Focus Areas:

- Definition and need of DSS
- Components (Data, Model, UI)
- Types of DSS
- Real-world applications (banking, healthcare)
- Advantages & limitations

Topic 2: Operational Systems vs DSS

☞ Focus Areas:

- OLTP vs OLAP concepts
- Key differences (data, users, purpose)
- Performance comparison
- Use-case scenarios
- Integration between systems

Topic 3: Architecture of Data Warehouse

☞ Focus Areas:

- Data warehouse definition
- Three-tier architecture
- ETL process
- Data marts & metadata
- Benefits in decision-making

Topic 4: OLAP and Data Cubes

☞ Focus Areas:

- OLAP concept
- Data cube structure
- OLAP operations (slice, dice, roll-up, drill-down)
- Applications in analytics

Topic 5: Dimensional Data Modeling

☞ Focus Areas:

- Fact and dimension tables
- Star schema vs snowflake schema
- Schema design process
- Real-world example

Topic 6: Query Processing & BI Tools

☞ Focus Areas:

- Query processing stages
- Query optimization
- BI tools overview
- Reporting and visualization
- Examples of open-source BI tools

Topic 7: OLAP Application using KNIME

☞ Focus Areas:

- Introduction to KNIME
- Workflow components (nodes, connections)
- Performing OLAP operations
- Sample application/demo
- Report generation

Code of Conduct

I M. Charan (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Charan

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

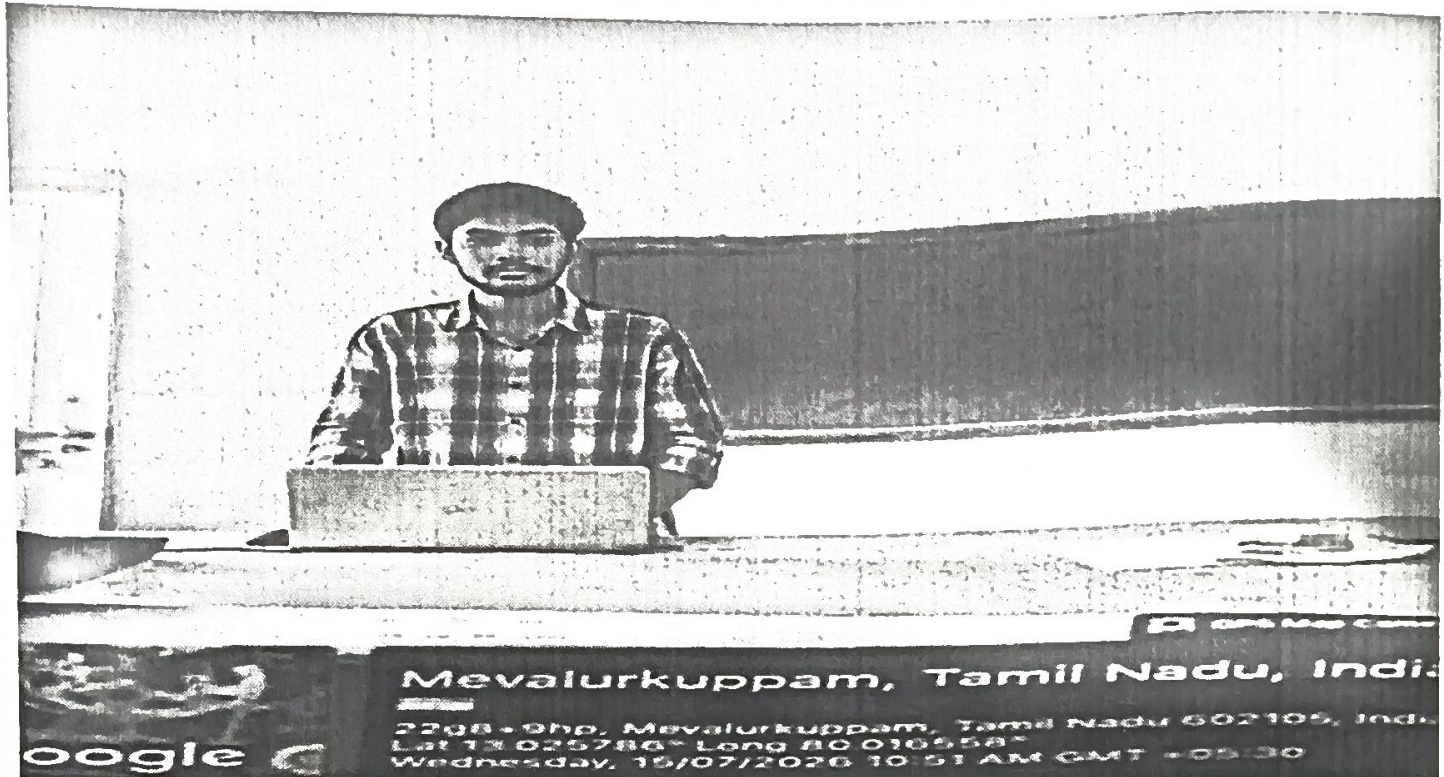
Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

CSA1610 - DATAWAREHOUSING AND DATAMINING

CO1-ASSESSMENT TOOL-2-TECHNICAL PRESENTATION

Topic 6: Query Processing & BI Tools



1. Query Processing Stages

Query processing is the step-by-step activity performed by a Database Management System (DBMS) to translate a high-level user query (like SQL) into a low-level execution plan.

- **Parsing and Translation:**
 - Checks the query syntax for errors.
 - Verifies table and column names against the database catalog.
 - Translates the SQL query into a relational algebra expression (an abstract syntax tree).
- **Optimization:**
 - Evaluates different ways to execute the query.
 - Selects the most efficient plan based on cost estimates (CPU, I/O time).
- **Code Generation / Compilation:**
 - Converts the chosen optimal execution plan into executable code or machine instructions.
- **Execution:**
 - The database runtime engine runs the generated code against the database files.
 - Fetches the data and returns the final results to the user.

2. Query Optimization

Query optimization is the process of finding the most efficient way to execute a query out of many possible alternatives.

- **Cost-Based Optimization (CBO):**
 - Uses database statistics (e.g., number of rows, index distributions, data size).
 - Calculates an estimated cost in terms of disk I/O and CPU usage for each plan.
 - Chooses the plan with the lowest estimated cost.
- **Rule-Based Optimization (RBO):**
 - Uses a fixed set of predefined rules or heuristics to choose an execution path.
 - *Example rule:* "Always use an index if one is available, regardless of table size." (Largely obsolete in modern systems).
- **Key Techniques:**
 - **Selection Pushdown:** Moving filters (like WHERE age > 21) deeper into the execution tree so rows are filtered out as early as possible.
 - **Join Ordering:** Arranging the order in which tables are joined to minimize intermediate result sizes.

3. BI Tools Overview

Business Intelligence (BI) tools are software applications used to ingest, transform, analyze, and present large volumes of structured and unstructured data.

- **Core Purpose:** Convert raw data into actionable insights for strategic decision-making.
- **Key Capabilities:**
 - Data integration from multiple sources (Data Warehouses, Cloud storage, API feeds).
 - Data preparation and cleaning (ETL/ELT processing).
 - Ad-hoc analysis and querying without requiring SQL expertise.
 - Real-time or scheduled automated data refreshing.

4. Reporting and Visualization

This phase focuses on translating data insights into human-readable, visual, and interactive formats.

- **Reporting:**

- Static or Interactive summaries of historical data (e.g., monthly sales reports, financial statements).
 - Often structured as tabular data or structured summaries.
 - Visualization:
 - The graphical representation of information using visual elements like charts, graphs, and maps.
 - Common Components:
 - Dashboards: Single-page screens hosting a collection of live visual metrics and Key Performance Indicators (KPIs).
 - Charts: Bar charts (for comparisons), Line charts (for trends over time), Pie charts (for compositions), and Heat maps (for density patterns).
 - Drill-Down / Slice-and-Dice: Interactive features allowing users to click a metric to see deeper, granular details.
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5. Examples of Open-Source BI Tools

Open-source BI tools offer robust analytics and data visualization capabilities without the expensive licensing fees of proprietary software like Microsoft Power BI or Tableau.

- Apache Superset: A modern, cloud-native data exploration and visualization platform that handles petabyte-scale data and integrates seamlessly with SQL databases.
- Metabase: A highly user-friendly BI tool that allows non-technical users to ask questions and build dashboards without knowing SQL.
- Grafana: While heavily used for infrastructure monitoring, it is an exceptionally powerful open-source dashboard tool for time-series data and analytics.
- KNIME: An open-source data analytics, reporting, and integration platform built on a visual workflow interface (excellent for advanced data mining and machine learning tasks).